

Design Delivery Package

Case t4_r1 — battery for extreme-cold environment equipment

Numbering scheme: VBF-T4R1-<DOC>-<NN> | Generated 2026-08-25

Signature block (left blank for manual signing)

Prepared	Reviewed	Approved

File list

File name	Number	Format	Source description
design_spec.md	VBF-T4R1-DS-01	md	design_spec generated per deliverable-design-spec spec; values from parameter set / cell/r6_t2_*.json
design_spec.pdf	VBF-T4R1-DS-01	pdf	PDF release of design_spec.md (reportlab)
bom.xlsx	VBF-T4R1-BOM-01	xlsx	bill of materials, dual caliber g/cell + kg/kWh (openpyxl); literature-default mass fractions annotated
bom.pdf	VBF-T4R1-BOM-01	pdf	PDF release of bom.xlsx (reportlab)
datasheet.md	VBF-T4R1-DSH-01	md	technical datasheet per deliverable-datasheet spec; simulation values only
datasheet.pdf	VBF-T4R1-DSH-01	pdf	PDF release of datasheet.md (reportlab)
calc.xlsx	VBF-T4R1-CALC-01	xlsx	design calculation sheet, 5 sheets: inputs / capacity-energy / energy density / N-P-mass / process (openpyxl)
calc.pdf	VBF-T4R1-CALC-01	pdf	PDF release of calc.xlsx (reportlab)
dvpr.md	VBF-T4R1-DVPR-01	md	design verification plan and report, virtual-test version per deliverable-dvpr spec
dvpr.pdf	VBF-T4R1-DVPR-01	pdf	PDF release of dvpr.md (reportlab)
dfmea.md	VBF-T4R1-DFMEA-01	md	qualitative design FMEA per deliverable-dfmea spec; simulation signals only
dfmea.pdf	VBF-T4R1-DFMEA-01	pdf	PDF release of dfmea.md (reportlab)
delivery_index.md	VBF-T4R1-IDX-01	md	delivery package index (this file)
delivery_index.pdf	VBF-T4R1-IDX-01	pdf	PDF release of delivery_index.md (reportlab, blueprint cover)